

Multiple Choice Questions

Q1. A hospital implements an AI system to prioritize patients in the emergency room. During a system audit, the engineers cannot explain why the algorithm prioritized a low-risk case over a critical one. Which Responsible AI principle is violated?

- . A. Fairness
- . B. Reliability and Safety
- . C. Transparency
- . D. Inclusivity

Correct Answer: C. Transparency

Explanation: Transparency requires that AI systems operate traceably and understandably so that humans can comprehend how specific decisions or predictions are made.

Q2. An automated financial trading bot performs perfectly during stable market conditions but causes massive, erratic losses during a brief, unpredicted network fluctuation. Which principle of Responsible AI does this system fail to meet?

- . A. Accountability
- . B. Reliability and Safety
- . C. Privacy and Security
- . D. Fairness

Correct Answer: B. Reliability and Safety

Explanation: Systems must operate reliably and safely under both normal conditions and unexpected edge cases or technical fluctuations.

Q3. A company deploys an AI facial recognition system for office entry, but

the system consistently fails to authenticate employees who use wheelchairs because the camera angles are fixed for standing heights. Which principle is violated?

- . A. Transparency
- . B. Accountability
- . C. Inclusivity
- . D. Privacy and Security

Correct Answer: C. Inclusivity

Explanation: Inclusivity dictates that AI systems must accommodate and empower people of all physical capabilities and backgrounds without creating physical or digital barriers.

Q4. A conversational AI application designed for legal assistance accidentally leaks confidential client

case files into a public database during a system updates transfer. Which principle has been compromised?

- . A. Fairness
- . B. Privacy and Security
- . C. Accountability
- . D. Transparency

Correct Answer: B. Privacy and Security

Explanation: AI systems must align with strict data protection laws to ensure user privacy and secure sensitive assets from unauthorized exposure.

Q5. When a state-sponsored automated driving system causes a collision, the software vendor claims the software cannot be held liable, while the vehicle manufacturer blames the radar hardware. No entity accepts ownership

of the system's behavior. Which principle is missing?

- . A. Fairness
- . B. Inclusivity
- . C. Accountability
- . D. Transparency

Correct Answer: C. Accountability

Explanation: Accountability establishes that the human designers, developers, and operators remain completely responsible for the ultimate actions and impacts of their AI systems.

Q6. A credit-scoring AI model systematically assigns lower credit scores to applicants from a specific minority demographic group, despite them having identical income levels to

other applicants. Which principle is violated?

- . A. Reliability and Safety
- . B. Fairness
- . C. Transparency
- . D. Accountability

Correct Answer: B. Fairness *Explanation:*

Fairness ensures that models do not discriminate or display systemic bias based on demographic features such as race, gender, or origin.

Q7. You need to explain the fundamental difference between general Artificial Intelligence and Machine Learning to a stakeholder. Which statement accurately describes Machine Learning?

- . A. It is a system that mimics human consciousness using rule-based hardcoded logic.
- . B. It is a subset of AI that uses historical data to train statistical models for future predictions without explicit programming.
- . C. It is a hardware structure built exclusively to store large language models.
- . D. It is an automated software architecture that writes its own legal frameworks.

Correct Answer: B. It is a subset of AI that uses historical data to train statistical models for future predictions without explicit programming.

Explanation: ML relies on data inputs to learn pattern behaviors organically rather

than following rigid, manually written code statements.

Q8. A deep learning architecture differs from traditional machine learning algorithms primarily through its core reliance on which structural component?

- . A. Relational SQL Tables
- . B. Rule-based decision loops
- . C. Artificial Neural Networks
- . D. Linear regression formulas

Correct Answer: C. Artificial Neural Networks *Explanation:* Deep Learning is a specialized subfield of ML that leverages multi-layered artificial neural networks to identify highly complex relationships in data.

Q9. You want to manage your data assets, computational models, logs, and target deployments from a single, centralized cloud environment in Azure. What asset must you create first?

- . A. Azure AI Bot Service
- . B. Azure ML Workspace
- . C. Inference Cluster
- . D. Compute Instance

Correct Answer: B. Azure ML Workspace

Explanation: The Azure Machine Learning Workspace acts as the foundational, top-level cloud resource to manage all experimental and operational artifacts.

Q10. A business analyst possesses a large dataset of historical customer churn details but has zero programming experience. They need to find the

absolute most accurate classification model quickly. Which Azure ML capability matches their requirement?

- . A. Azure ML Notebooks
- . B. Automated ML (AutoML)
- . C. Inference Clusters
- . D. Attached Compute

Correct Answer: B. Automated ML

(AutoML) *Explanation:* Automated ML sweeps across multiple machine learning algorithms, scaling hyperparameter adjustments automatically to locate the optimal model configuration without manual coding.

Q11. A developer wants to build a custom price forecasting model by visually linking data cleaning steps, normalization algorithms, and model

**testing modules on a dynamic canvas.
Which interface should they use?**

- . A. Azure ML Designer
- . B. Azure AI Language
- . C. Compute Clusters
- . D. Azure AI Document Intelligence

Correct Answer: A. Azure ML Designer

Explanation: Azure ML Designer provides an interactive, visual drag-and-drop workspace to construct end-to-end machine learning workflows without programmatic code.

Q12. You are using an Azure ML Workspace to write advanced custom Python training scripts using deep learning frameworks like PyTorch. Which resource provides the developer workstation environment?

- . A. Compute Cluster
- . B. Compute Instance
- . C. Inference Cluster
- . D. Attached Compute

Correct Answer: B. Compute Instance

Explanation: Compute Instances serve as fully configured, cloud-hosted development workstations targeted for data scientists writing custom scripts in Jupyter environments.

Q13. You have a massive training workload involving millions of rows of unstructured web logs that requires automated scaling across multiple virtual machines during training. Which Azure ML compute target should you allocate?

- . A. Compute Instance

- . B. Attached Compute
- . C. Compute Cluster
- . D. Inference Cluster

Correct Answer: C. Compute Cluster

Explanation: Compute Clusters are multi-node compute resources capable of scaling up or down automatically based on the computational demands of heavy training workloads.

Q14. You have successfully trained an optical character recognition model. You now want to deploy this model as a production-grade live API endpoint to process incoming requests from thousands of mobile users simultaneously. Which compute target should you use?

- . A. Compute Instance

- . B. Compute Cluster
- . C. Inference Cluster
- . D. Azure Databricks

Correct Answer: C. Inference Cluster

Explanation: Inference Clusters are optimized architectures tailored specifically to deploy trained machine learning models into production for low-latency, real-time application usage.

Q15. An agricultural drone application needs to process aerial photographs to determine if a crop field contains wheat, corn, or soy. It only needs to declare the primary crop type for each individual photo. Which computer vision category does this match?

- . A. Object Detection
- . B. Image Classification

- . C. Optical Character Recognition (OCR)
- . D. Named Entity Recognition

Correct Answer: B. Image Classification

Explanation: Image Classification assigns a single categorical label or tag to an entire input photograph based on its primary visual subject.

Q16. A security system scans real-time traffic camera video streams to isolate individual pedestrians, bicycles, and delivery vans, highlighting each item with a distinct visual box to track movements. Which technique is being applied?

- . A. Image Classification
- . B. Object Detection
- . C. Face Verification
- . D. Document Intelligence

Correct Answer: B. Object Detection

Explanation: Object Detection locates individual targets within a graphic frame, cataloging them and specifying their spatial coordinates using bounding boxes.

Q17. A mobile application allows users to point their device camera at a printed book page or hand-drawn whiteboards and instantly extract the visible text into an editable digital document file. Which specific technology is at work here?

- . A. Image Classification
- . B. Face Analysis
- . C. Optical Character Recognition (OCR)
- . D. Sentiment Analysis

Correct Answer: C. Optical Character Recognition (OCR) *Explanation:* OCR focuses exclusively on discovering,

isolating, and converting physical handwritten or printed text assets inside images into digital text characters.

Q18. A boutique fashion brand manufactures specialized custom hand-woven bags. They want to train an AI model to recognize their brand's specific products rather than generic backpacks or purses. Which Azure AI service should they leverage?

- . A. Azure AI Face
- . B. Azure AI Document Intelligence
- . C. Azure Custom Vision
- . D. Azure AI Language

Correct Answer: C. Azure Custom Vision

Explanation: Azure Custom Vision lets developers adapt predefined image models by supplying their own specialized dataset

to recognize distinct, custom domain classes.

Q19. An airport security checkpoint uses a camera system to read structural passports, extract passport numbers, dates of birth, and nationality strings directly into an organized registration database table. Which service is optimal?

- . A. Azure AI Vision (OCR)
- . B. Azure Custom Vision
- . C. Azure AI Document Intelligence
- . D. Azure AI Bot Service

Correct Answer: C. Azure AI Document Intelligence *Explanation:* Azure AI Document Intelligence incorporates advanced machine learning to parse structured text layouts, pulling out key-value

pairs, names, values, and tabular objects seamlessly.

Q20. An application scans thousands of online restaurant reviews written by customers and generates a metric score showing the percentage of users who expressed a highly negative view of the service. Which NLP feature is utilized?

- . A. Named Entity Recognition (NER)
- . B. Sentiment Analysis
- . C. Key Phrase Extraction
- . D. Speech-to-Text

Correct Answer: B. Sentiment Analysis

Explanation: Sentiment Analysis processes text arrays to categorize emotional tone, scoring evaluations across positive, neutral, or negative metrics.

Q21. A research tool evaluates medical essays to instantly flag recurring technical terms like "diabetes," "metformin," or "cardiology," classifying them as medical conditions or drugs. Which feature handles this task?

- . A. Key Phrase Extraction
- . B. Sentiment Analysis
- . C. Named Entity Recognition (NER)
- . D. Text-to-Speech

Correct Answer: C. Named Entity Recognition (NER) *Explanation:* NER uncovers specific entity classes (like individuals, locations, times, or custom domain nouns) inside unstructured text sequences.

Q22. A news aggregator application processes lengthy multi-page

journalistic articles to generate a compact list of the absolute most critical concepts and phrases discussed. Which service capability is required?

- . A. Named Entity Recognition (NER)
- . B. Key Phrase Extraction
- . C. Text-to-Speech
- . D. Azure AI Bot Service

Correct Answer: B. Key Phrase

Extraction *Explanation:* Key Phrase Extraction extracts primary conversation tokens or critical discussion highlights directly from large chunks of text data.

Q23. An accessibility option on an educational platform reads written text modules out loud to visually impaired students, ensuring the synthetic voice

sounds completely natural and human-like. Which service delivers this?

- . A. Speech-to-Text
- . B. Text-to-Speech
- . C. Azure AI Translation
- . D. Sentiment Analysis

Correct Answer: B. Text-to-Speech

Explanation: Text-to-Speech translates digital character text arrays into highly lifelike, artificial audio streams.

Q24. A voicemail application records voice messages left on an enterprise office line and delivers a typed summary of the spoken words directly to the employee's email inbox. Which technology is required?

- . A. Text-to-Speech
- . B. Speech-to-Text

- . C. Sentiment Analysis
- . D. Named Entity Recognition

Correct Answer: B. Speech-to-Text

Explanation: Speech-to-Text processes audible frequencies and maps them accurately into structured written words.

Q25. An international customer service platform instantly converts inbound English chat messages into clear German text responses so regional customer support agents can respond accurately. Which service handles this?

- . A. Azure AI Language
- . B. Azure AI Translation
- . C. Azure AI Speech
- . D. Azure ML Designer

Correct Answer: B. Azure AI Translation

Explanation: Azure AI Translation converts

multi-language textual documents or strings across separate localized language syntaxes in real time.

Q26. A digital bank wants to build an AI assistant within its mobile app that interacts with users via text, interprets questions regarding checking accounts, and resolves basic card issues autonomously. Which framework is designed for this?

- . A. Azure AI Translation
- . B. Azure Custom Vision
- . C. Azure AI Bot Service
- . D. Compute Instances

Correct Answer: C. Azure AI Bot Service

Explanation: Azure AI Bot Service supplies the direct enterprise ecosystem to curate, scale, and integrate multi-turn

conversational agents across digital consumer systems.

Q27. A machine learning project aims to classify customer reviews. The development team modifies an input database to utilize an external custom spark engine linked straight to their Azure ML Workspace for data processing. This external asset is known as:

- . A. Inference Cluster
- . B. Compute Instance
- . C. Attached Compute
- . D. Compute Cluster

Correct Answer: C. Attached Compute

Explanation: Attached Compute refers to linking pre-existing external compute architectures (like Azure Databricks or

HDInsight clusters) directly into an Azure ML Workspace environment.

Q28. During model deployment, a team decides to host a real-time web endpoint within a lightweight testing phase that does not require production scale or auto-scaling VM power. Which compute asset could they use for low-demand local testing?

- . A. Inference Cluster
- . B. Compute Instance
- . C. Compute Cluster
- . D. Attached Compute

Correct Answer: B. Compute Instance

Explanation: While Inference Clusters are best for high-scale production environments, a Compute Instance can act

as a local endpoint target for small-scale validation and development testing.

Q29. A legal firm wants to scan contracts and automatically hide all individual names, phone numbers, and home addresses before publishing the files. Which Azure AI Language feature helps locate these specific terms?

- . A. Key Phrase Extraction
- . B. Named Entity Recognition (NER)
- . C. Sentiment Analysis
- . D. Text-to-Speech

Correct Answer: B. Named Entity Recognition (NER) *Explanation:* NER identifies and categorizes specific tokens representing unique structural classifications like people, entities, data types, locations, and personal identifiers.

Q30. You are configuring an Automated ML run in Azure ML. What must you supply to the system so it can begin generating and testing models independently?

- . A. The final pre-trained weights file
- . B. The raw training dataset and the target column label to predict
- . C. A complex custom Python neural network script
- . D. A pre-built custom vision bounding box coordinates profile

Correct Answer: B. The raw training dataset and the target column label to predict *Explanation:* AutoML requires you to upload the dataset, select the task type (e.g., classification, regression), and specify the target feature column you intend to predict.

Q31. An AI system used by an insurance company systematically calculates higher premiums for individuals who live in low-income neighborhoods, regardless of their actual personal safety record. Which principle is violated?

- . A. Inclusivity
- . B. Accountability
- . C. Fairness
- . D. Transparency

Correct Answer: C. Fairness *Explanation:* Selecting systemic proxy values that result in biased treatments toward marginalized socioeconomic groups directly breaches the principle of Fairness.

Q32. A automated diagnostic chatbot gives random, conflicting medical

advice when users frame their health questions using non-standard dialect phrases. Which principle is violated?

- . A. Privacy and Security
- . B. Reliability and Safety
- . C. Accountability
- . D. Transparency

Correct Answer: B. Reliability and Safety

Explanation: Reliability ensures that unpredictable variations in input formats or user dialects do not cause the AI to generate erratic or unsafe outputs.

Q33. A public transit app deploys an AI navigation tool that completely ignores routing paths containing wheelchair ramps or elevators. Which principle must the engineering team fix?

- . A. Transparency

- . B. Fairness
- . C. Inclusivity
- . D. Accountability

Correct Answer: C. Inclusivity

Explanation: AI products must purposefully include design options that extend utility across diverse groups, including individuals with structural accessibility requirements.

Q34. A automated algorithm decides to deny an individual's small business loan request, but the output file displays only a cryptic error code with no mathematical explanation for the refusal. Which principle is violated?

- . A. Fairness
- . B. Transparency
- . C. Inclusivity
- . D. Reliability and Safety

Correct Answer: B. Transparency

Explanation: Transparency requires models to provide explainable paths behind high-stakes decisions so users can understand why an action occurred.

Q35. A social media analytics app uses an AI system to analyze video clips and count the total number of distinct faces that appear in crowds without storing individual biometric profiles. Which Azure service supports this?

- . A. Azure AI Document Intelligence
- . B. Azure AI Face
- . C. Azure AI Translation
- . D. Azure AI Bot Service

Correct Answer: B. Azure AI Face

Explanation: Azure AI Face provides face detection features to recognize, locate, and

count human faces within digital images or video frames.

Q36. You want to scan physical store receipts to extract the exact item descriptions, tax values, and purchase times into a corporate accounting tool. Which service is optimal?

- . A. Azure Custom Vision
- . B. Azure AI Document Intelligence
- . C. Azure AI Speech
- . D. Azure AI Translation

Correct Answer: B. Azure AI Document Intelligence *Explanation:* Document Intelligence uses tailored structural intelligence to identify specific receipt values, transforming document prints into machine-readable tabular structures.

Q37. A user states: "The customer service at this branch was completely unacceptable." An AI pipeline scans this text and flags it with a negative rating. Which feature was utilized?

- . A. Named Entity Recognition
- . B. Key Phrase Extraction
- . C. Sentiment Analysis
- . D. Text-to-Speech

Correct Answer: C. Sentiment Analysis

Explanation: Sentiment Analysis determines the underlying emotional sentiment (positive, negative, neutral) of a given text string.

Q38. An application needs to analyze legal documents to find all mentions of specific target companies, regulatory

laws, and active dates. Which feature should you select?

- . A. Key Phrase Extraction
- . B. Named Entity Recognition (NER)
- . C. Sentiment Analysis
- . D. Speech-to-Text

Correct Answer: B. Named Entity Recognition (NER) *Explanation:* NER isolates critical nouns and tags them according to set classifications (e.g., Organizations, Regulations, Dates).

Q39. A software system reads a textbook document out loud to a student, adjusting the pacing and inflection to sound identical to a professional human narrator. This is powered by:

- . A. Speech-to-Text
- . B. Text-to-Speech

- . C. Sentiment Analysis
- . D. Azure AI Translation

Correct Answer: B. Text-to-Speech

Explanation: Text-to-Speech converts written text scripts into high-quality, audible human speech outputs.

Q40. A mobile translation app allows two individuals speaking completely different languages to communicate by instantly converting spoken sentences from one language into the other. Which combination of services supports this?

- . A. Azure AI Speech and Azure AI Translation
- . B. Azure Custom Vision and Azure ML Designer
- . C. Azure AI Document Intelligence and Azure AI Face

- . D. Azure ML Notebooks and Compute Instances

Correct Answer: A. Azure AI Speech and Azure AI Translation *Explanation:* The app must convert spoken words to text (Speech), translate that text into another language (Translation), and optionally speak it out loud (Speech).

Q41. A customer support platform wants to build an automated agent that can answer frequent shipping questions on their web portal, escalating complex tickets to human support teams when necessary. Which service handles this?

- . A. Azure Custom Vision
- . B. Azure AI Translation
- . C. Azure AI Bot Service
- . D. Compute Clusters

Correct Answer: C. Azure AI Bot Service

Explanation: Azure AI Bot Service provides the foundation to build interactive, conversational agents that can chat with customers and handle basic automated workflows.

Q42. You are using Azure ML Designer to construct an automated data pipeline. You want to clean missing values and partition data into training sets visually. How are these operations represented on the screen?

- . A. As custom Python code scripts
- . B. As visual drag-and-drop modules connected by lines
- . C. As rows in a relational database table
- . D. As bounding boxes on an image

Correct Answer: B. As visual drag-and-drop modules connected by lines

Explanation: Azure ML Designer uses a visual canvas where distinct data processing steps and algorithms are represented as modules that you connect to form a pipeline flowchart.

Q43. A developer wants to build a predictive model but does not know which algorithm (e.g., Random Forest vs. Support Vector Machine) will yield the lowest error rate. Which Azure ML feature can automatically resolve this?

- . A. Compute Instances
- . B. Automated ML (AutoML)
- . C. Azure AI Face
- . D. Azure AI Translation

Correct Answer: B. Automated ML

(AutoML) *Explanation:* AutoML tests a variety of machine learning algorithms and hyperparameter combinations automatically to find the most accurate model for your data.

Q44. A security app uses cameras to detect when a vehicle parks illegally in a restricted zone. The app must draw a square box around the offending vehicle on the monitor. This is an example of:

- . A. Image Classification
- . B. Object Detection
- . C. Optical Character Recognition (OCR)
- . D. Key Phrase Extraction

Correct Answer: B. Object Detection

Explanation: Object detection identifies specific items within an image and

determines their exact location using bounding boxes.

Q45. An organization wants to ensure that all AI models deployed across its network are monitored by a specific human governance board to track systemic updates and take responsibility for any algorithmic errors. Which principle does this validate?

- . A. Fairness
- . B. Inclusivity
- . C. Accountability
- . D. Transparency

Correct Answer: C. Accountability

Explanation: Accountability ensures that human teams remain responsible for the oversight, behavior, and real-world impacts of deployed AI systems.